SIS SAMPLE DRAFT — FULL TIER

Westchester County Sample — Shopping Center / Retail

Traffic Impact Study

Prepared August 18, 2026 · 41.0180, -73.8000

SIS sample draft — full tier

Screening estimate — not for design submittal without independent verification by a licensed PE. | See /legal/disclaimer.

Traffic Impact Study

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TRAFFIC IMPACT STUDY FOR DOT PROJECTS

Prepared in Accordance With Chapter 5 of the NYSDOT Highway Design Manual (HDM)

Project name	Westchester County Sample — Shopping Center / Retail
PIN	PIN xxxx.xx (to be assigned by NYSDOT)
Project description	Shopping Center / Retail (85 1,000 sqft GLA) within New York-Newark-Jersey City MSA.
NYSDOT Region	Region 8 — Hudson Valley
Site coordinates	41.0180°, -73.8000°
Opening year (ETC)	2027

Note:
It is a violation of law for any person, unless they are acting under the direction of a licensed professional engineer, architect, landscape architect, or land surveyor, to alter an item in any way. If an item bearing the stamp of a licensed professional is altered, the altering engineer, architect, landscape architect, or land surveyor shall stamp the document and include the notation "altered by" followed by their signature, the date of such alteration, and a specific description of the alteration.

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

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PROJECT CONTEXT

NYSDOT region: Region 8 — Hudson Valley
Planning group: NYMTC (New York Metropolitan Transportation Council)
Regulatory regime: SEQRA (ECL Article 8 / 6 NYCRR Part 617).
SEQRA classification: Not yet determined — lead agency to classify as Type I (6 NYCRR Part 617.4 — likely EIS), Type II (Part 617.5 — exempt), or Unlisted (Part 617.6 — Short EAF + significance finding) before locking the TIS.

1.0 SUMMARY OF TRAFFIC IMPACTS

Capacity:

The proposed project will maintain an acceptable Level of Service of D or better in design year 2027.

Crash:

A minimum three-year accident history review was performed and the safety screening met all of the required steps. A full crash analysis is not required.

GML §239 referral: no state or county classified roadway was identified within 500 ft of the site via the NYSDOT RDM FeatureServer. The lead agency should still verify against the county's locally-adopted §239 mapping (some counties extend referral to additional facilities — e.g. county-designated scenic byways).

9	0	0	2.4s
INTERSECTIONS	LOS DROPS	AT LOS E/F	WORST Δ DELAY

2.0 TRAVEL SPEEDS

Per NYSDOT HDM Chapter 5 §5.2, posted speed limits and actual operating speeds (85th-percentile) along the study network are reported below. Posted-speed-limit values in the third column are auto-ingested from the NYSDOT Roadway Data Management (RDM) FeatureServer (https://gis.dot.ny.gov/hostingny/rest/services/Roadways/RDM_Roadway_Current/FeatureServer) at PDF-generation time — the same source data viewable through the NYSDOT Traffic Data Viewer (https://gis.dot.ny.gov/html5viewer/?viewer=tdv). Operating speeds (85th-percentile) are NOT published by NYSDOT and require a project-specific radar / floating-car speed study.

Street	Functional class	Posted speed	85th-pctile operating speed
South Central Avenue	Urban Minor Arterial	30 mph	Speed study required
Near East Hartsdale Avenue (E HARTSDALE AVE)	Urban Minor Arterial	Not posted (local)	Speed study required
East Hartsdale Avenue	Refer to NYSDOT TDV	—	Speed study required
Central Park Avenue	Urban Principal Arterial Other	30 mph	Speed study required
Walworth Avenue (WALWORTH AVENUE)	Urban Major Collector	30 mph	Speed study required
Fenimore Road (WALWORTH AVENUE)	Urban Major Collector	30 mph	Speed study required

3.0 CAPACITY ANALYSIS

Capacity analyses performed in this report are consistent with HCM 6th Edition. The software used to perform this analysis is the SimpleImpactStudies HCM 6 solver (per-approach control-delay model). Synchro / SimTraffic, HCS, Sidra, or VISSIM may be substituted for formal submittal per NYSDOT Regional Traffic Office preference.

Study network functional-class composition (per NYSDOT RDM): Urban Principal Arterial Other (4); Urban Minor Arterial (2); Urban Major Collector (2). NYSDOT functional class drives access management expectations under HDM Appendix 5A (entrance standards for state highways) and the default K-factor banding under HDM §5.3.

Level of Service thresholds applied below are per HCM 2010 Exhibits 18-4 (signalized), 19-1 / 20-2 / 21-1 (unsignalized — 2-way stop / all-way stop / roundabout), and 10-7 (freeway). Any component v/c > 1.0 is LOS F regardless of delay or density.

Signalized intersections (HCM 2010 Exhibit 18-4, p. 18-6) — control delay sec/veh; any v/c > 1.0 is F.

LOS	A	B	C	D	E	F
sec/veh	≤10	10–20	20–35	35–55	55–80	>80

Unsignalized — 2-way stop / all-way stop / roundabout (HCM 2010 Exhibits 19-1, 20-2, 21-1) — sec/veh; any v/c > 1.0 is F.

LOS	A	B	C	D	E	F
sec/veh	≤10	10–15	15–25	25–35	35–50	>50

Freeway basic / weaving / merge-diverge (HCM 2010 Exhibit 10-7, p. 10-9) — density pc/mi/ln; any component vd/c > 1.00 is F.

LOS	A	B	C	D	E	F
pc/mi/ln	≤11	11–18	18–26	26–35	35–45	>45

3.1 Growth Rates

Background traffic growth is applied at 1.5% per year over 1 year. This rate is the per-station median CAGR measured from the NYSDOT Traffic Monitoring AADT FeatureServer for Region 8 — Hudson Valley: 0.07%/yr median across 12,244 count stations (rolling actual-count window approximately 2000-2019; 25th-75th percentile range -1.57% to 1.80%). The measured value supersedes the NYSDOT statewide default fallbacks (per HDM Chapter 5: 1.7%/yr Interstate + ramps from the Data Services Bureau, 2.0%/yr all other roads from the Regional Planning Group). The controlling Regional Planning Group NYMTC (New York Metropolitan Transportation Council) may have separately-adopted growth-rate guidance that overrides the measured median for formal submittal — confirm at scoping.

3.2 Existing AADT and DHV

Existing AADT is sourced from the NYSDOT Traffic Data Viewer. Design Hourly Volume (DHV) is reported per HDM Chapter 5 convention as AADT × K-factor; this analysis applies K = 0.10 as the statewide design default — facility-specific K should be substituted from NYSDOT count-station records for formal submittal. The table below grows the existing volume at 1.5%/yr to Estimated

Time of Completion (ETC), ETC + 20 years, and ETC + 30 years per HDM Chapter 5 §5.3.

Intersection	2026 AADT	2026 DHV	2027 (ETC) AADT	2027 (ETC) DHV	2047 AADT	2047 DHV	2057 AADT	2057 DHV
South Central Avenue & West Hartsdale Avenue	11,790	1,179	11,967	1,197	16,118	1,612	18,705	1,871
Near East Hartsdale Avenue	6,130	613	6,222	622	8,380	838	9,725	973
East Hartsdale Avenue & Pipeline Road	6,130	613	6,222	622	8,380	838	9,725	973
Central Park Avenue & North Central Avenue	9,550	955	9,693	969	13,055	1,306	15,151	1,515
Walworth Avenue & Bronx River Parkway	1,230	123	1,248	125	1,681	168	1,951	195
Fenimore Road & Walworth Avenue	4,429	443	4,495	450	6,055	605	7,027	703
North Central Avenue & Central Park Avenue	9,551	955	9,694	969	13,057	1,306	15,153	1,515
Central Park Avenue & North Central Avenue	9,550	955	9,693	969	13,055	1,306	15,151	1,515
Battle Avenue & Central Avenue	10,600	1,060	10,759	1,076	14,491	1,449	16,817	1,682

Source: AADT estimated from engine peak-hour entering-volume sum via K = 0.10. NYSDOT Traffic Data Viewer measured AADT is recommended for submittal. Projection applies the 1.5%/yr background growth rate compounded annually.

3.3 Traffic Control Device Data

Existing traffic control devices at each study-network intersection are summarized below. Signal timing data (cycle length, phase splits, offset) and per-approach detection presence should be confirmed against the controlling NYSDOT Regional Traffic Office or municipal traffic-engineering signal record for formal submittal.

Intersection	Control type	Approaches w/ detection
South Central Avenue & West Hartsdale Avenue	Signal (controlling RTO to confirm)	Field verification required
Near East Hartsdale Avenue	Signal (controlling RTO to confirm)	Field verification required
East Hartsdale Avenue & Pipeline Road	Signal (controlling RTO to confirm)	Field verification required
Central Park Avenue & North Central Avenue	Signal (controlling RTO to confirm)	Field verification required
Walworth Avenue & Bronx River Parkway	Signal (controlling RTO to confirm)	Field verification required
Fenimore Road & Walworth Avenue	Signal (controlling RTO to confirm)	Field verification required
North Central Avenue & Central Park Avenue	Signal (controlling RTO to confirm)	Field verification required
Central Park Avenue & North Central Avenue	Signal (controlling RTO to confirm)	Field verification required
Battle Avenue & Central Avenue	Signal (controlling RTO to confirm)	Field verification required

3.4 Capacity Analysis for Existing No-Build Condition

Existing No-Build conditions reflect current-year peak-hour volumes (no growth, no project trips). Existing measured volumes are grown to opening year 2027 at 1.5%/yr to establish the No-Build baseline for §3.5 comparison.

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
South Central Avenue & West Hartsdale Avenue	C	23.4	C	23.7	0.66
Near East Hartsdale Avenue	B	17.2	B	17.3	0.35
East Hartsdale Avenue & Pipeline Road	B	17.2	B	17.3	0.35
Central Park Avenue & North Central Avenue	C	20.4	C	20.5	0.54
Walworth Avenue & Bronx River Parkway	B	14.2	B	14.2	0.07
Fenimore Road & Walworth Avenue	B	16.0	B	16.1	0.25
North Central Avenue & Central Park Avenue	C	20.4	C	20.5	0.54

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Central Park Avenue & North Central Avenue	C	20.4	C	20.5	0.54
Battle Avenue & Central Avenue	C	21.7	C	21.9	0.60

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

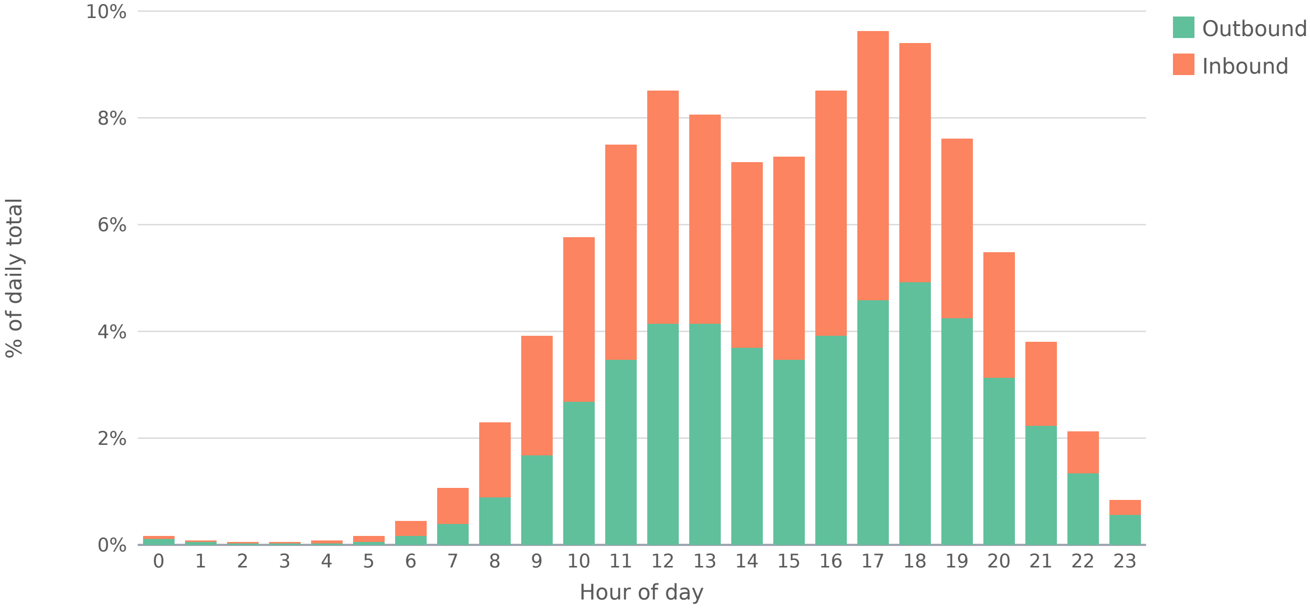
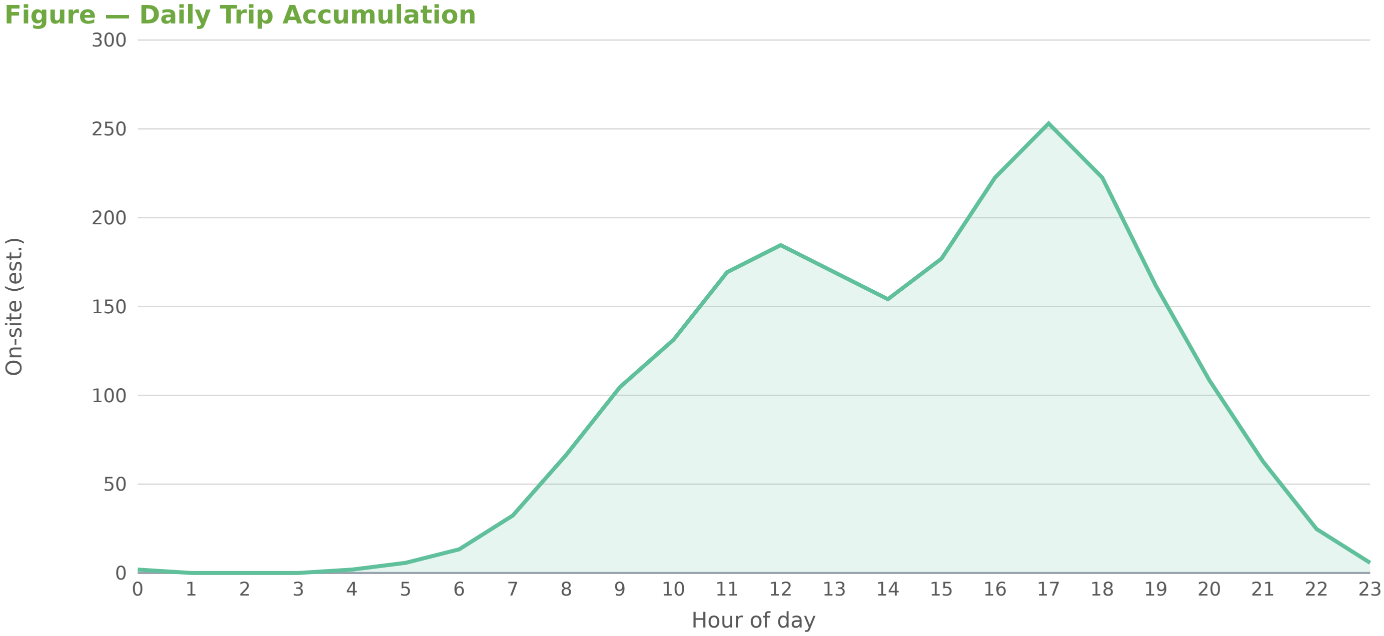


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

3.5 Capacity Analysis for Proposed Build Condition

Build conditions add the project's external peak-hour trips (680 PM peak, 326 inbound / 354 outbound) to the No-Build baseline at each affected intersection. Per HDM Chapter 5, intersection projects should also analyze each adjacent intersection plus any signal within ½ mile of the project site if expected left-turn volume changes materially affect capacity. The engine's network is determined by a 1.00-mile screening radius; manual scoping should confirm the ½-mile left-turn rule is satisfied.

Intersection	ETC No-Build	ETC Build	ETC+20 NB	ETC+20 Bld	Δ delay (s)
South Central Avenue & West Hartsdale Avenue	C	C	D	D	2.4
Near East Hartsdale Avenue	B	B	B	B	0.6
East Hartsdale Avenue & Pipeline Road	B	B	B	B	0.2
Central Park Avenue & North Central Avenue	C	C	C	C	0.2
Walworth Avenue & Bronx River Parkway	B	B	B	B	0.1
Fenimore Road & Walworth Avenue	B	B	B	B	0.0
North Central Avenue & Central Park Avenue	C	C	C	C	0.1
Central Park Avenue & North Central Avenue	C	C	C	C	0.1
Battle Avenue & Central Avenue	C	C	C	C	0.0

3.6 Capacity Improvement Measures

No capacity improvements are required under Build conditions. The project will operate within acceptable LOS standards at all affected intersections per HDM Chapter 5. Standard HDM Chapter 5 improvement categories (turn lanes, signal installation / re-timing, signing and striping, roundabout conversion, TDM / TSM per HDM Chapter 24) remain available if site-specific concerns arise during NYSDOT Regional Traffic Office review.

3.7 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = M / (d^1 · d_site); mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

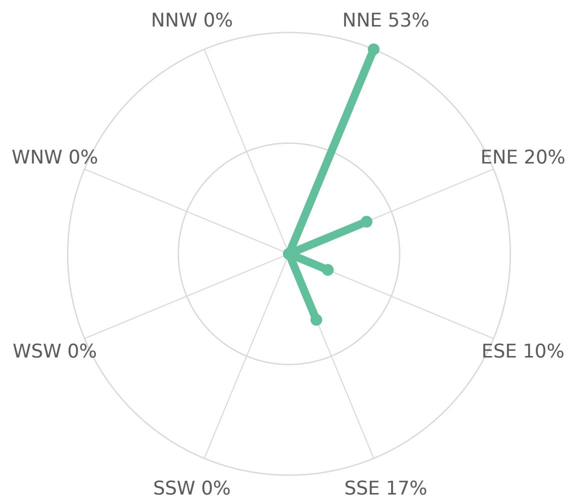
Directional sector	Share of project trips
Northeast (NNE + ENE)	73%
Southeast (ESE + SSE)	27%
Southwest (SSW + WSW)	0%
Northwest (WNW + NNW)	0%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
South Central Avenue & West Hartsdale Avenue	ENE	0.11	1,179	0.2	20.05%
Central Park Avenue & North Central Avenue	NNE	0.59	955	0.1	13.60%
Battle Avenue & Central Avenue	NNE	0.95	1,060	0.1	13.46%
North Central Avenue & Central Park Avenue	NNE	0.71	955	0.1	13.10%
Central Park Avenue & North Central Avenue	NNE	0.81	955	0.1	12.69%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Near East Hartsdale Avenue	ESE	0.19	613	0.1	10.04%
East Hartsdale Avenue & Pipeline Road	SSE	0.43	613	0.1	9.23%
Fenimore Road & Walworth Avenue	SSE	0.70	443	0.1	6.10%
Walworth Avenue & Bronx River Parkway	SSE	0.62	123	0.0	1.74%

9 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length ∝ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

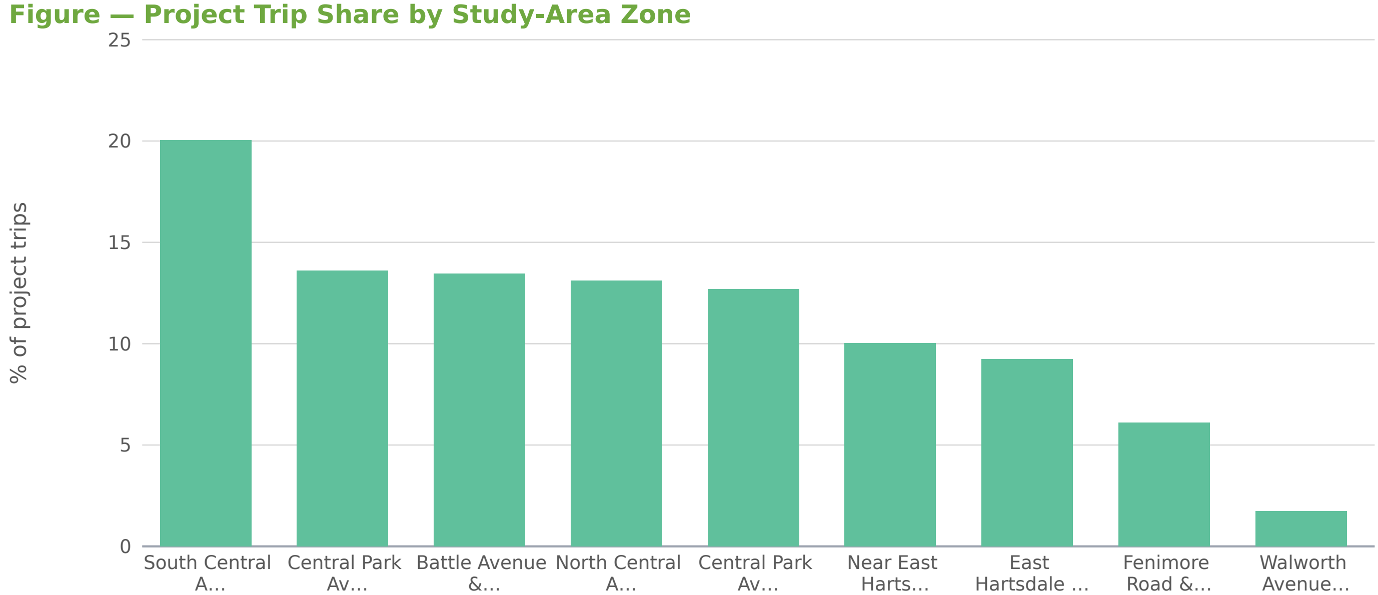
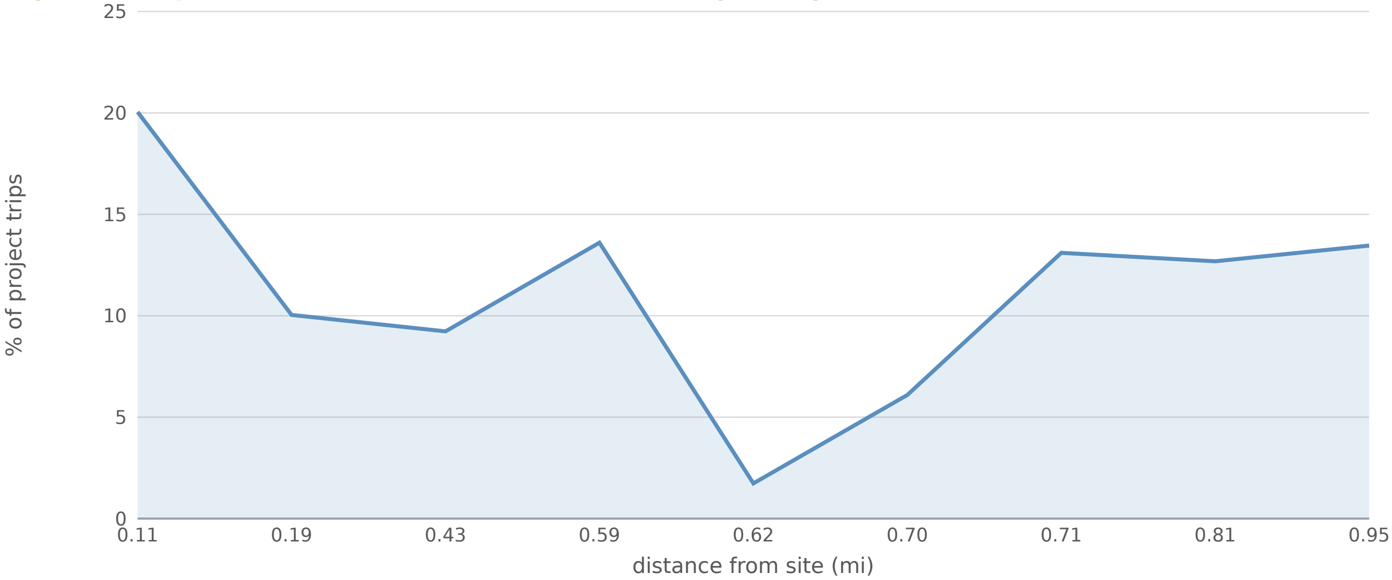


Figure — Trip Share vs. Distance from Site (Gravity Decay)



3.8 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
South Central Avenue & West Hartsdale Avenue	ENE	0.11	65	123	46.2%
Near East Hartsdale Avenue	ESE	0.19	42	80	30.1%
East Hartsdale Avenue & Pipeline Road	SSE	0.43	13	25	9.4%
Central Park Avenue & North Central Avenue	NNE	0.59	6	11	4.1%
Walworth Avenue & Bronx River Parkway	SSE	0.62	5	9	3.4%
Fenimore Road & Walworth Avenue	SSE	0.70	3	6	2.3%
North Central Avenue & Central Park Avenue	NNE	0.71	3	6	2.3%
Central Park Avenue & North Central Avenue	NNE	0.81	2	4	1.5%
Battle Avenue & Central Avenue	NNE	0.95	1	2	0.8%

Project trips assigned to each study intersection from the distribution shares above: the directional shares orient the loading toward the high-share sectors, distance-decayed to each approach.

4.0 CRASH ANALYSIS

A minimum three-year accident history review was performed for the period 2023 to 2026 per HDM Chapter 5 §5.3.4. County-level crash counts below are auto-ingested from the New York State Police "Motor Vehicle Crashes – Case Information: Three Year Window" Open Data dataset (https://data.ny.gov/Transportation/Motor-Vehicle-Crashes-Case-Information-Three-Year-/e8ky-4vqe), filtered by county_name resolved from the project site coordinates via the NYSDOT RDM_Roadway_Current FeatureServer. The values cover the entire rolling 3-year reporting window for WESTCHESTER County, statewide-system-wide — they are NOT a per-segment or per-intersection rate.

Severity descriptor (NY DMV)	Count (3-yr rolling)	Annualized

Severity descriptor (NY DMV)	Count (3-yr rolling)	Annualized
Fatal Accident	147	49.0
Injury Accident	3,622	1207.3
Property Damage Accident	63,027	21009.0
Property Damage & Injury Accident	16,543	5514.3
Total (all severities)	83,339	27779.7

Within WESTCHESTER County over the rolling 3-year window, 83,339 crashes were reported to NY DMV (147 fatal, 20,165 injury-involved, 63,027 property damage only). The Fatal share is 0.059% of all reported crashes (annualized). For the Safety Screening per HDM Chapter 5 §5.3.4, the project-specific rate must be normalized against the controlling facility's exposure (segment AADT × segment length; intersection AADT × entry-flow) for comparison to NYSDOT Office of Modal Safety statewide averages and the 1.5× HAL trigger. County-totals above provide a county-baseline reference only — site-radius (1/4-mile typical) crash records, HAL / PIL / SDL flags, and CMF / CRF mitigation values must be obtained from the Region 8 — Hudson Valley Regional Traffic Office via FOIL or SIMS. Required NYSDOT forms when the §4.0 section is expanded for formal submittal: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation), TE-204a (Accident Rate Summary), TE-213 (HAL / PIL / SDL Screening Output).

APPENDIX A — EXISTING VOLUME REPORT

Per-approach existing peak-hour volumes (vph) for all affected intersections within study limits. Source: SimpleImpactStudies HCM 6 solver, calibrated against the controlling DOT 511 / Regional Traffic Office data feed.

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
South Central Avenue & West Hartsdale Avenue	NB	366	371	371
South Central Avenue & West Hartsdale Avenue	SB	308	312	351
South Central Avenue & West Hartsdale Avenue	EB	273	277	341
South Central Avenue & West Hartsdale Avenue	WB	233	236	257
Near East Hartsdale Avenue	NB	198	201	216
Near East Hartsdale Avenue	SB	157	160	160
Near East Hartsdale Avenue	EB	124	126	168
Near East Hartsdale Avenue	WB	133	135	158
East Hartsdale Avenue & Pipeline Road	NB	178	181	189
East Hartsdale Avenue & Pipeline Road	SB	171	174	187
East Hartsdale Avenue & Pipeline Road	EB	141	143	143
East Hartsdale Avenue & Pipeline Road	WB	122	124	129
Central Park Avenue & North Central Avenue	NB	248	252	258
Central Park Avenue & North Central Avenue	SB	245	248	252
Central Park Avenue & North Central Avenue	EB	253	257	257
Central Park Avenue & North Central Avenue	WB	209	212	213
Walworth Avenue & Bronx River Parkway	NB	41	42	44
Walworth Avenue & Bronx River Parkway	SB	29	30	35
Walworth Avenue & Bronx River Parkway	EB	29	29	29
Walworth Avenue & Bronx River Parkway	WB	24	24	26
Fenimore Road & Walworth Avenue	NB	123	125	127
Fenimore Road & Walworth Avenue	SB	123	125	128
Fenimore Road & Walworth Avenue	EB	101	103	103
Fenimore Road & Walworth Avenue	WB	95	97	98
North Central Avenue & Central Park Avenue	NB	317	321	325
North Central Avenue & Central Park Avenue	SB	214	217	219
North Central Avenue & Central Park Avenue	EB	264	268	268
North Central Avenue & Central Park Avenue	WB	161	164	164
Central Park Avenue & North Central Avenue	NB	291	295	297
Central Park Avenue & North Central Avenue	SB	236	239	241
Central Park Avenue & North Central Avenue	EB	247	251	251
Central Park Avenue & North Central Avenue	WB	182	184	185
Battle Avenue & Central Avenue	NB	283	287	288
Battle Avenue & Central Avenue	SB	285	289	289
Battle Avenue & Central Avenue	EB	259	262	262

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Battle Avenue & Central Avenue	WB	235	238	238

APPENDIX B — EXISTING CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Existing No-Build condition (current-year volumes; volumes grown to opening year at the §3.1 background-growth rate for No-Build comparison).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
South Central Avenue & West Hartsdale Avenue	NB	371	0.46	19.0	B
South Central Avenue & West Hartsdale Avenue	SB	312	0.39	17.9	B
South Central Avenue & West Hartsdale Avenue	EB	277	0.34	17.2	B
South Central Avenue & West Hartsdale Avenue	WB	236	0.29	16.6	B
Near East Hartsdale Avenue	NB	201	0.25	16.1	B
Near East Hartsdale Avenue	SB	160	0.20	15.5	B
Near East Hartsdale Avenue	EB	126	0.16	15.0	B
Near East Hartsdale Avenue	WB	135	0.17	15.2	B
East Hartsdale Avenue & Pipeline Road	NB	181	0.22	15.8	B
East Hartsdale Avenue & Pipeline Road	SB	174	0.21	15.7	B
East Hartsdale Avenue & Pipeline Road	EB	143	0.18	15.3	B
East Hartsdale Avenue & Pipeline Road	WB	124	0.15	15.0	B
Central Park Avenue & North Central Avenue	NB	252	0.31	16.8	B
Central Park Avenue & North Central Avenue	SB	248	0.31	16.8	B
Central Park Avenue & North Central Avenue	EB	257	0.32	16.9	B
Central Park Avenue & North Central Avenue	WB	212	0.26	16.2	B
Walworth Avenue & Bronx River Parkway	NB	42	0.05	14.1	B
Walworth Avenue & Bronx River Parkway	SB	30	0.04	13.9	B
Walworth Avenue & Bronx River Parkway	EB	29	0.04	13.9	B
Walworth Avenue & Bronx River Parkway	WB	24	0.03	13.9	B
Fenimore Road & Walworth Avenue	NB	125	0.15	15.0	B
Fenimore Road & Walworth Avenue	SB	125	0.15	15.0	B
Fenimore Road & Walworth Avenue	EB	103	0.13	14.8	B
Fenimore Road & Walworth Avenue	WB	97	0.12	14.7	B
North Central Avenue & Central Park Avenue	NB	321	0.40	18.0	B
North Central Avenue & Central Park Avenue	SB	217	0.27	16.3	B
North Central Avenue & Central Park Avenue	EB	268	0.33	17.1	B
North Central Avenue & Central Park Avenue	WB	164	0.20	15.5	B
Central Park Avenue & North Central Avenue	NB	295	0.36	17.6	B
Central Park Avenue & North Central Avenue	SB	239	0.30	16.6	B
Central Park Avenue & North Central Avenue	EB	251	0.31	16.8	B
Central Park Avenue & North Central Avenue	WB	184	0.23	15.8	B
Battle Avenue & Central Avenue	NB	287	0.35	17.4	B
Battle Avenue & Central Avenue	SB	289	0.36	17.4	B
Battle Avenue & Central Avenue	EB	262	0.32	17.0	B
Battle Avenue & Central Avenue	WB	238	0.29	16.6	B

APPENDIX C — PROPOSED CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Proposed Build condition (No-Build volumes plus project external trips at the assigned distribution).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
South Central Avenue & West Hartsdale Avenue	NB	371	0.46	19.0	B
South Central Avenue & West Hartsdale Avenue	SB	351	0.43	18.6	B
South Central Avenue & West Hartsdale Avenue	EB	341	0.42	18.4	B
South Central Avenue & West Hartsdale Avenue	WB	257	0.32	16.9	B
Near East Hartsdale Avenue	NB	216	0.27	16.3	B
Near East Hartsdale Avenue	SB	160	0.20	15.5	B
Near East Hartsdale Avenue	EB	168	0.21	15.6	B
Near East Hartsdale Avenue	WB	158	0.20	15.5	B
East Hartsdale Avenue & Pipeline Road	NB	189	0.23	15.9	B
East Hartsdale Avenue & Pipeline Road	SB	187	0.23	15.9	B
East Hartsdale Avenue & Pipeline Road	EB	143	0.18	15.3	B
East Hartsdale Avenue & Pipeline Road	WB	129	0.16	15.1	B
Central Park Avenue & North Central Avenue	NB	258	0.32	16.9	B
Central Park Avenue & North Central Avenue	SB	252	0.31	16.8	B
Central Park Avenue & North Central Avenue	EB	257	0.32	16.9	B
Central Park Avenue & North Central Avenue	WB	213	0.26	16.2	B
Walworth Avenue & Bronx River Parkway	NB	44	0.05	14.1	B
Walworth Avenue & Bronx River Parkway	SB	35	0.04	14.0	B
Walworth Avenue & Bronx River Parkway	EB	29	0.04	13.9	B
Walworth Avenue & Bronx River Parkway	WB	26	0.03	13.9	B
Fenimore Road & Walworth Avenue	NB	127	0.16	15.1	B
Fenimore Road & Walworth Avenue	SB	128	0.16	15.1	B
Fenimore Road & Walworth Avenue	EB	103	0.13	14.8	B
Fenimore Road & Walworth Avenue	WB	98	0.12	14.7	B
North Central Avenue & Central Park Avenue	NB	325	0.40	18.1	B
North Central Avenue & Central Park Avenue	SB	219	0.27	16.3	B
North Central Avenue & Central Park Avenue	EB	268	0.33	17.1	B
North Central Avenue & Central Park Avenue	WB	164	0.20	15.5	B
Central Park Avenue & North Central Avenue	NB	297	0.37	17.6	B
Central Park Avenue & North Central Avenue	SB	241	0.30	16.6	B
Central Park Avenue & North Central Avenue	EB	251	0.31	16.8	B
Central Park Avenue & North Central Avenue	WB	185	0.23	15.8	B
Battle Avenue & Central Avenue	NB	288	0.36	17.4	B
Battle Avenue & Central Avenue	SB	289	0.36	17.5	B
Battle Avenue & Central Avenue	EB	262	0.32	17.0	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Battle Avenue & Central Avenue	WB	238	0.29	16.6	B

APPENDIX D — CRASH ANALYSIS DIAGRAMS AND TABLES

Crash analysis diagrams and tables are not produced by this screening analysis. When the §4.0 crash-analysis section is expanded for formal submittal, the following NYSDOT forms are required: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation Form), TE-204a (Accident Rate Summary), and TE-213 (HAL / PIL / SDL Screening Output). SIMS output and statewide-average rate references should be obtained from the controlling Regional Traffic Office per HDM Chapter 5 §5.3.4 and the NYSDOT Office of Modal Safety statewide rate tables (https://www.dot.ny.gov/divisions/operating/osss/highway/accident-rates).

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \Sigma (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: a=28507, b=-0.02, c=-0.123) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
South Central Avenue & West Hartsdale Avenue	0.11	0.3	123	46.2%
Near East Hartsdale Avenue	0.19	0.5	80	30.1%
East Hartsdale Avenue & Pipeline Road	0.43	1.0	25	9.4%
Central Park Avenue & North Central Avenue	0.59	1.4	11	4.1%
Walworth Avenue & Bronx River Parkway	0.62	1.5	9	3.4%
Fenimore Road & Walworth Avenue	0.70	1.7	6	2.3%
North Central Avenue & Central Park Avenue	0.71	1.7	6	2.3%
Central Park Avenue & North Central Avenue	0.81	1.9	4	1.5%
Battle Avenue & Central Avenue	0.95	2.3	2	0.8%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 44% applied via a binary logit $P(auto) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 56% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 0.73. Per-signal v/c, delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

Screening estimate — not for design submittal without independent verification by a licensed PE. | See /legal/disclaimer.

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

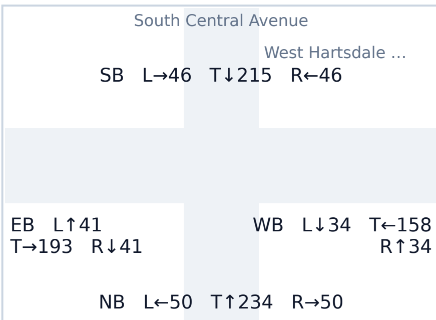
Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 South Central Avenue & West Hartsdale Avenue

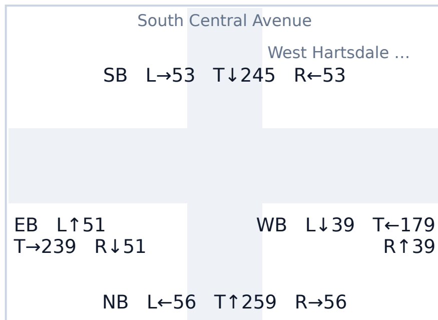
Signal ID	new-york-11889
Location	NE New York-Newark-Jersey City · 0.11 mi from site
Added PM peak trips	123
Existing / No-Build (PM)	LOS C · 23.7 s/veh · v/c 0.66
Build (PM)	LOS C · 26.1 s/veh · v/c 0.73
Δ control delay	+2.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

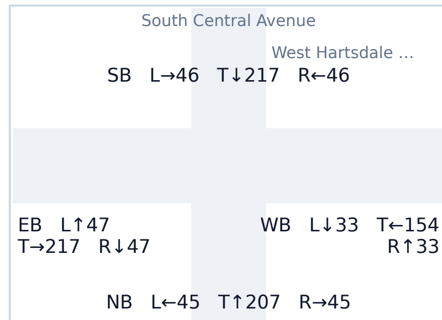
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	371	0	371	0.46 → 0.46	19.0 → 19.0	B → B	265
SB	312	39	351	0.39 → 0.43	17.9 → 18.6	B → B	247
EB	277	64	341	0.34 → 0.42	17.2 → 18.4	B → B	238
WB	236	20	257	0.29 → 0.32	16.6 → 16.9	B → B	170

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Left	42	34%
SB Right	39	32%
EB Through	22	18%
WB Through	20	16%

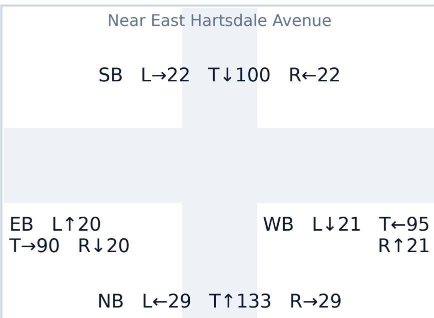
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Near East Hartsdale Avenue

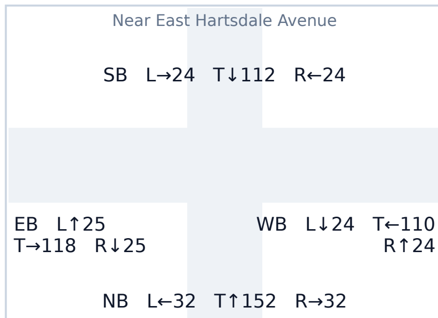
Signal ID	new-york-12277
Location	NE New York-Newark-Jersey City · 0.19 mi from site
Added PM peak trips	80
Existing / No-Build (PM)	LOS B · 17.3 s/veh · v/c 0.35
Build (PM)	LOS B · 17.9 s/veh · v/c 0.39
Δ control delay	+0.6 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

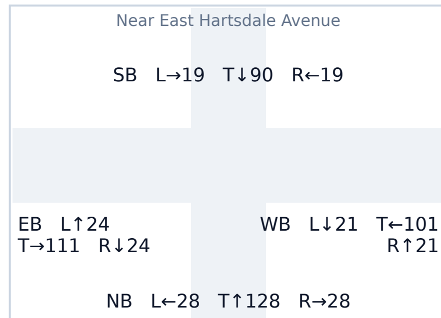
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	201	15	216	0.25 → 0.27	16.1 → 16.3	B → B	139
SB	160	0	160	0.20 → 0.20	15.5 → 15.5	B → B	99
EB	126	42	168	0.16 → 0.21	15.0 → 15.6	B → B	105
WB	135	23	158	0.17 → 0.20	15.2 → 15.5	B → B	98

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	25	31%
WB Through	23	29%
EB Right	17	21%
NB Left	15	19%

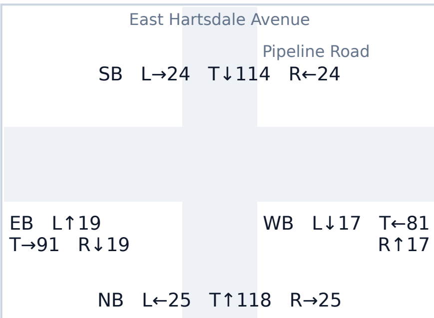
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 East Hartsdale Avenue & Pipeline Road

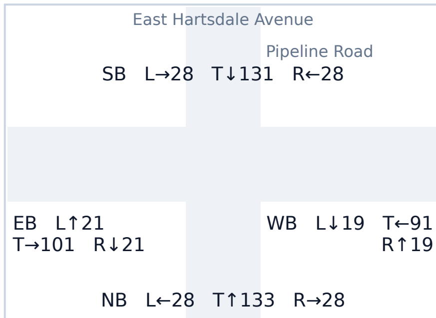
Signal ID	new-york-14260
Location	NE New York-Newark-Jersey City · 0.43 mi from site
Added PM peak trips	25
Existing / No-Build (PM)	LOS B · 17.3 s/veh · v/c 0.35
Build (PM)	LOS B · 17.5 s/veh · v/c 0.36
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

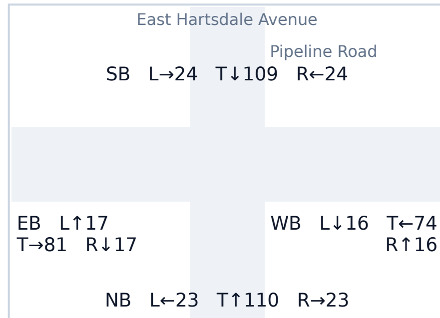
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	181	8	189	0.22 → 0.23	15.8 → 15.9	B → B	119
SB	174	13	187	0.21 → 0.23	15.7 → 15.9	B → B	118
EB	143	0	143	0.18 → 0.18	15.3 → 15.3	B → B	88
WB	124	4	129	0.15 → 0.16	15.0 → 15.1	B → B	79

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	8	32%
NB Through	8	32%
SB Left	5	20%
WB Right	4	16%

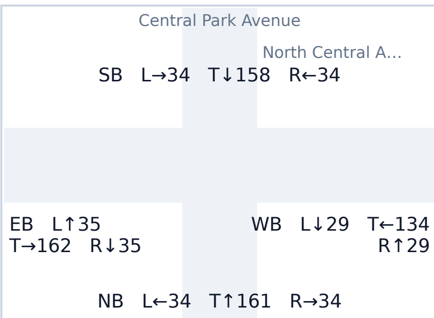
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Central Park Avenue & North Central Avenue

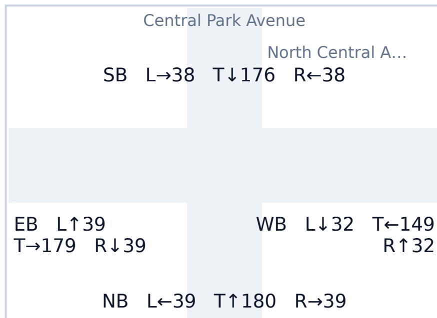
Signal ID	new-york-11890
Location	NE New York-Newark-Jersey City · 0.59 mi from site
Added PM peak trips	11
Existing / No-Build (PM)	LOS C · 20.5 s/veh · v/c 0.54
Build (PM)	LOS C · 20.7 s/veh · v/c 0.54
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

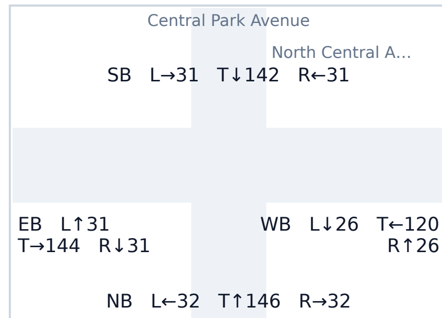
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	252	6	258	0.31 → 0.32	16.8 → 16.9	B → B	171
SB	248	4	252	0.31 → 0.31	16.8 → 16.8	B → B	166
EB	257	0	257	0.32 → 0.32	16.9 → 16.9	B → B	170
WB	212	1	213	0.26 → 0.26	16.2 → 16.2	B → B	137

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	4	36%
SB Through	4	36%
NB Right	2	18%
WB Left	1	9%

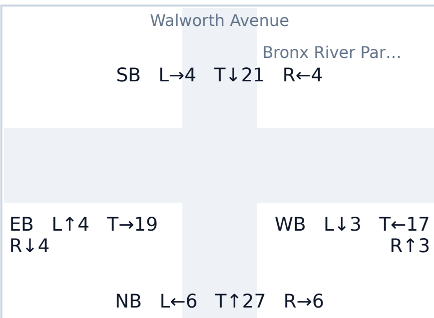
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Walworth Avenue & Bronx River Parkway

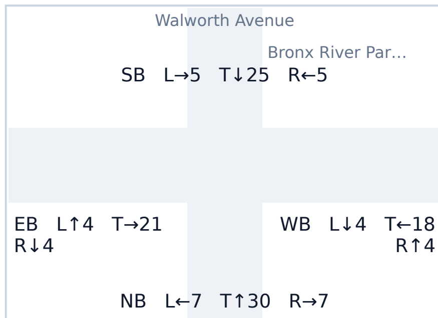
Signal ID	new-york-16985
Location	NE New York-Newark-Jersey City · 0.62 mi from site
Added PM peak trips	9
Existing / No-Build (PM)	LOS B · 14.2 s/veh · v/c 0.07
Build (PM)	LOS B · 14.3 s/veh · v/c 0.07
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

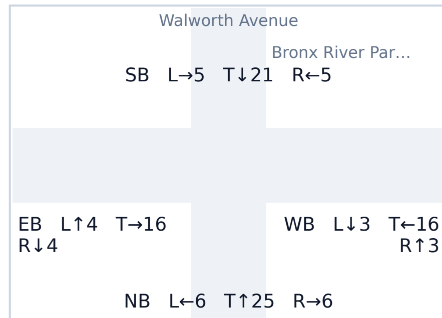
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	42	2	44	0.05 → 0.05	14.1 → 14.1	B → B	26
SB	30	5	35	0.04 → 0.04	13.9 → 14.0	B → B	20
EB	29	0	29	0.04 → 0.04	13.9 → 13.9	B → B	17
WB	24	2	26	0.03 → 0.03	13.9 → 13.9	B → B	15

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	3	33%
NB Through	2	22%
SB Left	2	22%
WB Right	2	22%

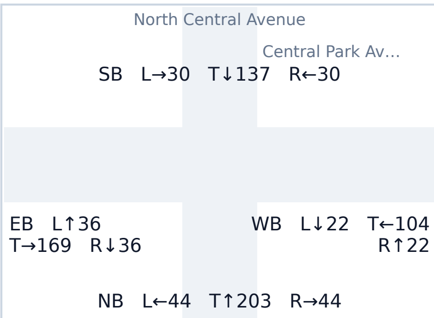
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 North Central Avenue & Central Park Avenue

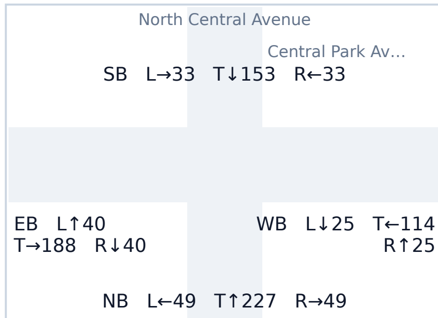
Signal ID	new-york-18291
Location	NE New York-Newark-Jersey City · 0.71 mi from site
Added PM peak trips	6
Existing / No-Build (PM)	LOS C · 20.5 s/veh · v/c 0.54
Build (PM)	LOS C · 20.6 s/veh · v/c 0.54
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

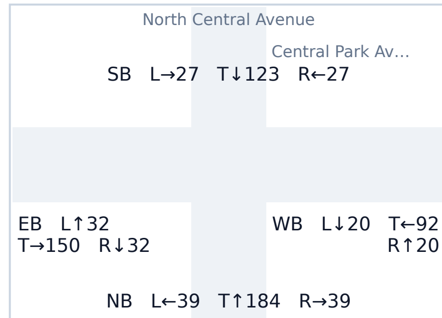
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	321	3	325	0.40 → 0.40	18.0 → 18.1	B → B	225
SB	217	2	219	0.27 → 0.27	16.3 → 16.3	B → B	141
EB	268	0	268	0.33 → 0.33	17.1 → 17.1	B → B	178
WB	164	1	164	0.20 → 0.20	15.5 → 15.5	B → B	103

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	2	33%
SB Through	2	33%
NB Right	1	17%
WB Left	1	17%

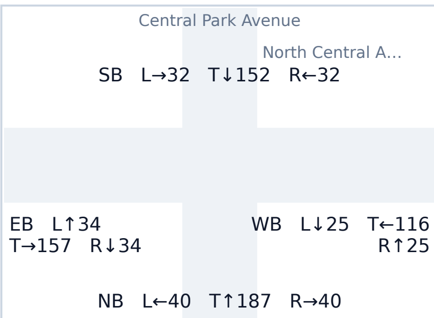
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Central Park Avenue & North Central Avenue

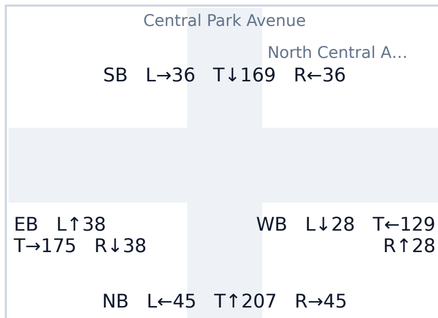
Signal ID	new-york-20116
Location	NE New York-Newark-Jersey City · 0.81 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS C · 20.5 s/veh · v/c 0.54
Build (PM)	LOS C · 20.6 s/veh · v/c 0.54
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

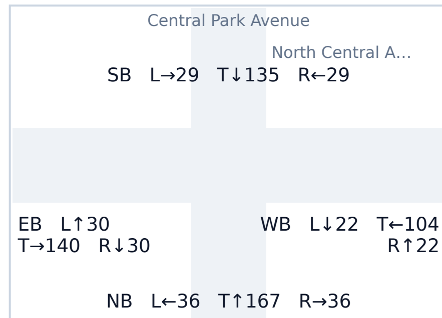
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	295	2	297	0.36 → 0.37	17.6 → 17.6	B → B	202
SB	239	1	241	0.30 → 0.30	16.6 → 16.6	B → B	157
EB	251	0	251	0.31 → 0.31	16.8 → 16.8	B → B	165
WB	184	1	185	0.23 → 0.23	15.8 → 15.8	B → B	117

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	1	25%
SB Through	1	25%
NB Right	1	25%
WB Left	1	25%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

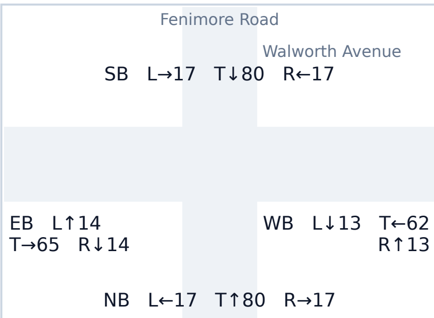
Screening estimate — not for design submittal without independent verification by a licensed PE. | See /legal/disclaimer.

A.8 Fenimore Road & Walworth Avenue

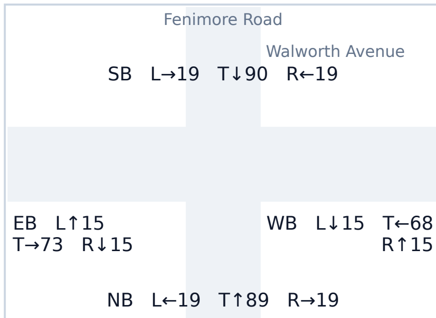
Signal ID	new-york-16984
Location	NE New York-Newark-Jersey City · 0.70 mi from site
Added PM peak trips	6
Existing / No-Build (PM)	LOS B · 16.1 s/veh · v/c 0.25
Build (PM)	LOS B · 16.1 s/veh · v/c 0.25
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

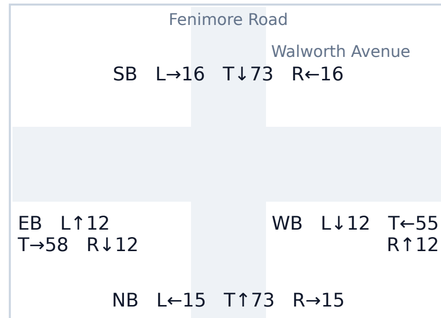
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	125	2	127	0.15 → 0.16	15.0 → 15.1	B → B	78
SB	125	3	128	0.15 → 0.16	15.0 → 15.1	B → B	78
EB	103	0	103	0.13 → 0.13	14.8 → 14.8	B → B	62
WB	97	1	98	0.12 → 0.12	14.7 → 14.7	B → B	59

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	2	33%
NB Through	2	33%
SB Left	1	17%
WB Right	1	17%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

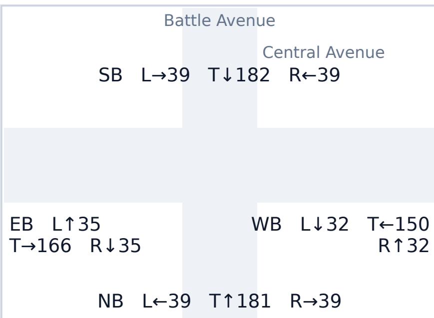
Screening estimate — not for design submittal without independent verification by a licensed PE. | See /legal/disclaimer.

A.9 Battle Avenue & Central Avenue

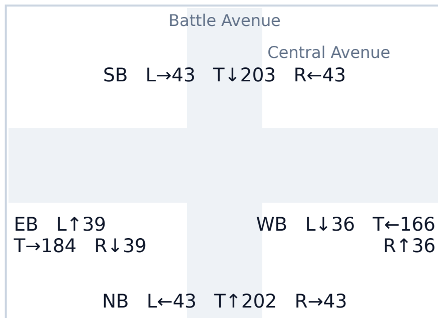
Signal ID	new-york-18287
Location	NE New York-Newark-Jersey City · 0.95 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS C · 21.9 s/veh · v/c 0.60
Build (PM)	LOS C · 21.9 s/veh · v/c 0.60
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

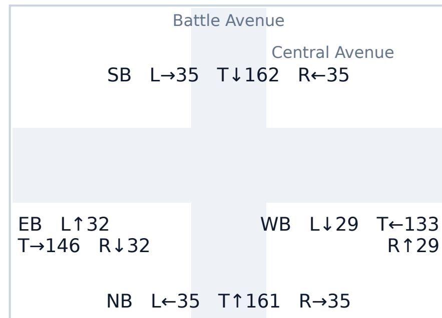
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	287	1	288	0.35 → 0.36	17.4 → 17.4	B → B	194
SB	289	1	289	0.36 → 0.36	17.4 → 17.5	B → B	196
EB	262	0	262	0.32 → 0.32	17.0 → 17.0	B → B	174
WB	238	0	238	0.29 → 0.29	16.6 → 16.6	B → B	156

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	1	50%
SB Through	1	50%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.